

STAT 7650: Computational Statistics

6. Convex Optimization

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Spring 2026

Outline

- 1 Convex optimization problems
 - Linear programming
 - Quadratic programming
- 2 Duality
- 3 KKT conditions
- 4 Interior-point method
 - Barrier method
 - Primal-dual interior-point method

Reference: Boyd and Vandenberghe (2004). Chapter 4, 5, 11.

Optimization Problem with Constraints

Let $x \in \mathbb{R}^n$. Consider the problem,

$$\begin{aligned} & \text{minimize} && f_0(x) \\ & \text{subject to} && f_i(x) \leq 0, \quad i = 1, \dots, m \\ & && h_j(x) = 0, \quad j = 1, \dots, p \end{aligned}$$

A point x is **feasible** if it satisfies all the constraints. The **feasible set** $\mathcal{X}$ include all feasible points. The **optimal value** p^* is

$$p^* = \inf \{ f_0(x) \mid f_i(x) \leq 0, i = 1, \dots, m, h_j(x) = 0, j = 1, \dots, p \}$$

An **optimal point** x^* solves the problem if $f_0(x^*) = p^*$, and the **optimal set** contains all optimal points,

$$\mathcal{X}_{\text{opt}} = \{ x \mid f_i(x) \leq 0, i = 1, \dots, m, h_j(x) = 0, j = 1, \dots, p, f_0(x) = p^* \}$$

Affine Set and Convex Set

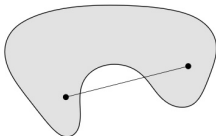
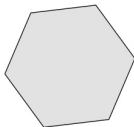
A set $C \subset \mathbb{R}^n$ is **affine** if the line through any two distinct points in C lies in C , that is, if for any $x_1, x_2 \in C$, we have

$$\theta x_1 + (1 - \theta)x_2 \in C, \quad \theta \in \mathbb{R}$$

It is a linear subspace dropping the “through the origin” requirement.

A set C is **convex** if the line segment between any two points in C lies in C , that is, if for any $x_1, x_2 \in C$, we have

$$\theta x_1 + (1 - \theta)x_2 \in C, \quad 0 \leq \theta \leq 1$$



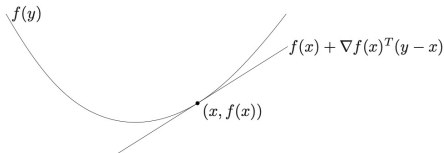
Convex Function

For a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, let $\text{dom}(f)$ denote the domain of f . A function f is **convex** if $\text{dom}(f)$ is a convex set and for any two arbitrary points $x, y \in \text{dom}(f)$, we have

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y), \quad 0 \leq \theta \leq 1$$

If f is differentiable, f is convex if and only if $\text{dom}(f)$ is convex and

$$f(y) \geq f(x) + \nabla f(x)^T(y - x), \quad x, y \in \text{dom}(f)$$



If f is twice differentiable, f is convex if and only if its Hessian matrix $\nabla^2 f(x)$ is positive semidefinite for any $x \in \text{dom}(f)$.

Convex Optimization

For a convex optimization problem,

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & a_j^T x = b_j, \quad j = 1, \dots, p\end{array}$$

where $f_0, f_1, \dots, f_m$ are convex functions.

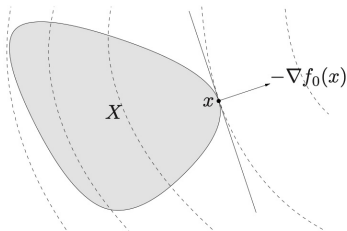
- The objective function f_0 must be convex
- The inequality constraint functions $f_1, \dots, f_m$ must be convex
- The equality constraint function must be affine

It can be shown that the feasible set $\mathcal{X}$ of this problem is convex.

Any locally optimal point for convex optimization is globally optimal.

Optimality Criterion

$x \in \mathcal{X}$ is optimal if and only if $\nabla f_0(x)^T(y - x) \geq 0$ for all $y \in \mathcal{X}$.



For an unconstrained problem, the condition reduces to $\nabla f_0(x) = 0$.

Problem with Equality Constraints Only

With only equality constraints and no inequality constraints,

$$\min f_0(x), \quad \text{subject to } Ax = b.$$

The optimality condition reduces to

$$\nabla f_0(x) \perp \mathcal{N}(A), \quad \text{or equivalently} \quad \nabla f_0(x) \in \mathcal{R}(A^T)$$

where $\mathcal{N}(A) = \{v \mid Av = 0\}$ is the nullspace of A and $\mathcal{R}(A^T)$ is the row space of A . Hence, there exists a $v \in \mathbb{R}^p$ such that

$$\nabla f_0(x) + A^T v = 0$$

which is the Lagrange multiplier optimality condition.

Minimization over the Nonnegative Orthant

Consider

$$\min f_0(x), \quad \text{subject to } x \geq 0.$$

The optimality condition becomes

$$x \geq 0, \quad \nabla f_0(x)^T (y - x) \geq 0 \text{ for any } x, y \geq 0$$

which implies

$$x \geq 0, \quad \nabla f_0(x) \geq 0, \quad (\nabla f_0(x))_i x_i = 0, \quad i = 1, \dots, n.$$

Linear Programming

A linear programming program is

$$\min c^T x, \quad \text{subject to } Gx \leq h, \quad Ax = b.$$

A standard form LP is

$$\min c^T x, \quad \text{subject to } Ax = b, \quad x \geq 0.$$

Transform a general LP to the standard form,

- Introduce slack variables $s \geq 0$ such that $Gx + s = h$
- Express x as $x = x^+ - x^-$, where $x^+ \geq 0$ and $x^- \geq 0$

Quantile

Let $F(x)$ be the CDF of a random variable X . The τ th quantile is

$$F^{-1}(\tau) = \inf\{x : F(x) \geq \tau\}, \quad 0 < \tau < 1$$

The median is the 0.5th quantile. Consider

$$\min_x E[\rho_\tau(X - a)], \quad \text{where } \rho_\tau(u) = \begin{cases} u\tau, & u \geq 0 \\ -u(1 - \tau), & u < 0 \end{cases}$$

When the minimizer is unique, $\hat{a} = F^{-1}(\tau)$; otherwise, we have an interval of τ th quantiles.

When $\tau = 0.5$, the function $\rho_{0.5}(u) = 0.5|u|$.

Quantile Regression

The observations $\{(y_i, x_i), y_i \in \mathbb{R}, x_i \in \mathbb{R}^p, i = 1, \dots, n\}$ are independently sampled from a population with the τ th conditional quantile function as $Q_y(\tau \mid x) = \beta_\tau^T x$. We can estimate β by

$$\min_{\beta \in \mathbb{R}^p} \sum_{i=1}^n \rho_\tau(y_i - \beta^T x_i)$$

where $\rho_\tau(u) = u(\tau - I(u < 0))$. Introduce $r = y - X\beta$ and r^+, r^- ,

$$\min_{\beta, r^+, r^-} \frac{1}{n} (\tau r^+ + (1 - \tau) r^-)$$

$$\text{subject to } y - X\beta = r^+ - r^-, r^+ \geq 0, r^- \geq 0$$

When $\tau = 0.5$, the loss function is least absolute deviations (LAD).

Quadratic Programming

A **quadratic programming** problem extends linear programming by allowing a quadratic objective function.

Find an n -dimensional vector x that

$$\begin{aligned} & \text{minimize} && \frac{1}{2}x^T Qx + x^T c \\ & \text{subject to} && Ax \leq b \quad (\text{inequality constraints}) \\ & && Bx = d \quad (\text{equality constraints}) \\ & && \ell_i \leq x \leq u_i \quad (\text{bound constraints}) \end{aligned}$$

where Q is a semi-positive definite matrix.

Huber Loss and Robust Regression

The Huber loss function is defined as

$$H_M(u) = \begin{cases} u^2, & |u| \leq M \\ 2M|u| - M^2, & |u| > M \end{cases}$$

The robust regression with Huber loss is defined as

$$\min_{\beta} \frac{1}{2} \sum_{i=1}^n H_M(y_i - \beta^T x_i)$$

Split each residual into three parts: $r_i = y_i - \beta^T x_i = u_i - v_i + w_i$.

$$\min \frac{1}{2} \sum_{i=1}^n w_i^2 + M \sum_{i=1}^n (u_i + v_i)$$

subject to $y_i - \beta^T x_i = u_i - v_i + w_i$,

$$u_i \geq 0, v_i \geq 0, -M \leq w_i \leq M$$

Software

- CVXR (<https://cvxr.rbind.io/>): Convex optimization in R
- Gurobi (<https://www.gurobi.com/>): Free academic license.
Provide interface to most major programming languages.

Let $\beta = \beta^+ - \beta^-$, $\beta^+ \geq 0$, and $\beta^- \geq 0$. The lasso becomes

$$\min_{\beta^+, \beta^-} \frac{1}{2} \|y - X(\beta^+ - \beta^-)\|^2 + \lambda 1^T(\beta^+ + \beta^-)$$

or equivalently, the objective function is

$$\frac{1}{2} \begin{pmatrix} \beta^+ \\ \beta^- \end{pmatrix}^T \begin{pmatrix} X^T X & -X^T X \\ -X^T X & X^T X \end{pmatrix} \begin{pmatrix} \beta^+ \\ \beta^- \end{pmatrix} + \begin{pmatrix} \lambda 1 - X^T y \\ \lambda 1 + X^T y \end{pmatrix}^T \begin{pmatrix} \beta^+ \\ \beta^- \end{pmatrix} + \frac{1}{2} y^T y$$

```
library(CVXR)

# y: response vector, length = n
# x: design matrix, n-by-p matrix

n = NROW(x); p = NCOL(x)

bvar = Variable(p)
obj = sum_squares(y - x %*% bvar) / 2 + lambda * p_norm(bvar, 1)

prob = Problem(Minimize(obj))
result = solve(prob)

b = result$getValue(bvar)      # estimated coefficients
f_value = result$value         # min objective function
```

CVXR is convenient to use, and good for small-sized or moderate-sized problems.


```
library(gurobi);
n = NROW(x);  p = NCOL(x);  xtx = t(x) %*% x;  xty = t(x) %*% y;

model = list(modelsense = "min")

model$Q = 0.5 * rbind(cbind(xtx, -xtx), cbind(-xtx, xtx))
model$obj = c(lambda - xty, lambda + xty)
model$objcon = 0.5 * sum(y * y)
model$lb = rep(0.0, p + p)
model$ub = rep(Inf, p + p)

model$A = matrix(0.0, nrow = 0, ncol = p + p)
model$sense = rep(">", 0)
model$rhs = rep(0.0, 0)

params = list(OutputFlag = 0)
QPfit = gurobi(model, params)

b = QPfit$x[1:p] - QPfit$x[(p+1):(p+p)] # estimated coefficients
f_value = QPfit$objval # min objective function
```

Components in Gurobi Output

status	a string, OPTIMAL or other strings
objval	the objective value of the computed solution
x	the computed solution
slack	slack variable, one entry for each row of A 0 means that the constraint is tight
pi	dual variable, one entry for each row of A

Lagrangian and Lagrange Dual Function

Consider an optimization problem in the standard form

$$\min f_0(x), \quad \text{subject to } f_i(x) \leq 0, \quad h_j(x) = 0.$$

The Lagrangian is

$$L(x, \lambda, v) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{j=1}^p v_j h_j(x)$$

where λ_i and v_j are **Lagrange multipliers** or **dual variables**. The **Lagrange dual function** is

$$g(\lambda, v) = \inf_{x \in \mathcal{D}} L(x, \lambda, v)$$

Let $\mathcal{D}$ be the domain, and the optimal value be p^* .

For any $\lambda \geq 0$ and any v , we have $g(\lambda, v) \leq p^*$.

Comments

- $g(\lambda, v)$ is concave, even when the original problem is not convex.
- Find the best lower bound that can be obtained from the Lagrange dual function

$$\max_{\lambda, v} g(\lambda, v), \quad \text{subject to } \lambda \geq 0$$

This is called the **Lagrange dual problem**, while the original problem is called the **primal problem**.

- The **dual feasible** means a pair (λ, v) such that $\lambda \geq 0$ and $g(\lambda, v) > -\infty$.
- The solution of the dual problem (λ^*, v^*) is called the **dual optimal** or **optimal Lagrange multipliers**.

Weak Dual and Strong Dual

Denote the optimal value of the Lagrange dual problem as d^* .

$$d^* \leq p^*$$

- This property holds even if the original problem is not convex.
- This property is called **weak duality**. The difference $p^* - d^*$ is called the **optimality duality gap** of the original problem.
- The **strong duality** holds if $d^* = p^*$. It holds for
 - linear programming
 - quadratic programming
 - many convex optimization with $f_0, f_1, \dots, f_m$ convex and affine equality constraints.

Dual of Linear Programming

Consider a LP in standard form,

$$\min c^T x, \quad \text{subject to } Ax = b, \quad x \geq 0$$

The Lagrangian is

$$L(x, \lambda, v) = c^T x - \lambda^T x + v^T (Ax - b) = -b^T v + (A^T v - \lambda + c)^T x$$

The dual function is

$$g(\lambda, v) = \begin{cases} -b^T v, & \text{if } A^T v - \lambda + c = 0 \\ -\infty, & \text{otherwise} \end{cases}$$

The dual problem of LP is

$$\max -b^T v, \quad \text{subject to } A^T v + c \geq 0.$$

The dual of a standard form LP is an LP with only inequality constraints, and vice versa.

Dual of Lasso

Let $r = y - X\beta$, the Lagrangian of the lasso problem is

$$L(\beta, r, u) = \frac{1}{2}\|r\|_2^2 + \lambda\|\beta\|_1 + u^T(y - X\beta - r)$$

The dual function is

$$g(u) = \inf_{r, \beta} L(\beta, r, u) = \begin{cases} -\frac{1}{2}\|u\|_2^2 + u^T y, & \|X^T u\|_\infty \leq \lambda \\ -\infty, & \text{otherwise} \end{cases}$$

The infimum is achieved when $r = u$. The dual problem is

$$\min_u \|y - u\|_2^2, \quad \text{subject to } \|X^T u\|_\infty \leq \lambda$$

Polyhedron and Projection

A **polyhedron** is the intersection of a finite number of half-spaces and hyperplanes.

$$C = \{x \mid a_i^T x \leq b_i, i = 1, \dots, m, c_j^T x = d_j, j = 1, \dots, p\}$$

A bounded polyhedron is sometimes called a **polytope**.

The projection onto a polyhedron C is well-defined,

$$P_C(x) = \arg \min_{u \in C} \|x - u\|_2$$

Lasso Fit as Projection

For any X and $\lambda \geq 0$, the lasso fit $\hat{y} = X\hat{\beta}$ can be written as

$$\hat{y} = X\hat{\beta}(\lambda) = y - \hat{u} = (I - P_C)(y)$$

where C is the polyhedron $C = \{u \in \mathbb{R}^n : \|X^T u\|_\infty \leq \lambda\}$.

Let $\mathcal{A} = \{k : \hat{\beta}_k \neq 0\}$. The degrees of freedom for lasso is

$$\text{df}(X\hat{\beta}) = E[\text{rank}(X_{\mathcal{A}})].$$

When X has full column rank, $\text{df}(X\hat{\beta}) = E[|\mathcal{A}|]$, where $|\mathcal{A}|$ is the number of nonzero in $\hat{\beta}$.

KKT Optimality Conditions

For a **convex optimization problem**,

$$\min f_0(x), \quad \text{subject to } f_i(x) \leq 0, \quad h_j(x) = 0.$$

Assume that the functions $f_0, \dots, f_m, h_1, \dots, h_p$ are differentiable. Let x^* and (λ^*, v^*) be any primal and dual optimal points with zero duality gap. The **Karush-Kuhn-Tucker** (KKT) conditions include

$$\nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) + \sum_{j=1}^p v_j^* \nabla h_j(x^*) = 0$$

$$f_i(x^*) \leq 0, i = 1, \dots, m$$

$$h_j(x^*) = 0, j = 1, \dots, p$$

$$\lambda_i^* \geq 0, i = 1, \dots, m$$

$$\lambda_i^* f_i(x^*) = 0, i = 1, \dots, m$$

Comments

For any optimization problem with differentiable objective and constraint functions for which strong duality obtains, any pair of primal and dual optimal points must satisfy the KKT conditions.

- When the primal problem is convex, the KKT conditions are also sufficient for the points to be primal and dual optimal.
- In other words, if f_i are convex and h_i are affine, and $\tilde{x}$, $\tilde{\lambda}$, $\tilde{v}$ are any points that satisfy the KKT conditions, then $\tilde{x}$ and $(\tilde{\lambda}, \tilde{v})$ are primal and dual optimal, with zero duality gap.

KKT Conditions for Lasso

Let $S = X^T X$ and $r = X^T y$. The KKT conditions for lasso is

$$S\beta - r + \lambda 1_p - w^+ = 0$$

$$-S\beta + r + \lambda 1_p - w^- = 0$$

$$w_k^+ \beta_k^+ = 0, \quad w_k^- \beta_k^- = 0$$

$$\beta_k^+ \geq 0, \quad \beta_k^- \geq 0, \quad w_k^+ \geq 0, \quad w_k^- \geq 0$$

Suppose that at given λ , we can define the active set as

$$\mathcal{A} = \{k : \hat{\beta}_k \neq 0\}$$

Then

$$X_k^T (Y - X\hat{\beta}) = \lambda \text{sign}(\hat{\beta}_k), \quad k \in \mathcal{A}$$

$$|X_k^T (Y - X\hat{\beta})| < \lambda, \quad k \notin \mathcal{A}$$

Solution Path Algorithm

We can write out the lasso estimate as

$$\hat{\beta}_{\mathcal{A}} = (X_{\mathcal{A}}^T X_{\mathcal{A}})^{-1} (X_{\mathcal{A}}^T Y - \lambda \text{sign}(\hat{\beta}_{\mathcal{A}})), \quad \hat{\beta}_{\mathcal{A}^c} = 0.$$

which implies that $\hat{\beta}$ is piecewise linear in λ ,

$$\frac{d\hat{\beta}_{\mathcal{A}}}{d\lambda} = -(X_{\mathcal{A}}^T X_{\mathcal{A}})^{-1} \text{sign}(\hat{\beta}_{\mathcal{A}})$$

Idea of the solution path algorithm

- Compute the solution path for $\infty > \lambda_1 > \lambda_2 > \dots > \lambda_m = 0$
- At each λ_k , either some variables enter $\mathcal{A}$ or leave $\mathcal{A}$
- $\beta_{\mathcal{A}}$ is a line segment between $(\lambda_k, \lambda_{k+1})$
- When $\lambda = \infty$, $\hat{\beta} = 0$
- When $\lambda = 0$, $\hat{\beta}$ is the least squares estimate

Solution Path Algorithm

Let knots be values of λ where $\mathcal{A}$ changes.

- Start at $\lambda = \infty$, $\hat{\beta} = 0$, and $\mathcal{A} = \emptyset$.
- First knot: $\lambda_1 = \|X^T y\|_\infty$ and add $j_1 = \arg \max |X_j^T y|$ into $\mathcal{A}$.
- Next knot: assume that the previous knot is λ_k .
 - Compute $d_{\mathcal{A}} = (X_{\mathcal{A}}^T X_{\mathcal{A}})^{-1} \text{sign}(\hat{\beta}_{\mathcal{A}})$

$$\beta_{\mathcal{A}}(\lambda) = \beta_{\mathcal{A}}(\lambda_k) + (\lambda_k - \lambda)d_{\mathcal{A}}, \quad \beta_{\mathcal{A}^c}(\lambda) = 0$$

- One variable enters $\mathcal{A}$ if $j \notin \mathcal{A}$ and $|X_j^T(y - X\hat{\beta})| = \lambda$
- One variable leaves $\mathcal{A}$ if $j \in \mathcal{A}$ and $\hat{\beta}_j(\lambda) = 0$
- Set λ_{k+1} to be the largest λ such that $\lambda_{k+1} < \lambda$ and $\mathcal{A}$ changes

Interior-Point Method

Consider a convex optimization problem,

$$\min f_0(x), \quad \text{subject to } f_i(x) \leq 0, \quad Ax = b.$$

Assume $f_0, \dots, f_m$ are convex and differentiable, $\text{rank}(A) = p < n$.

$$\min f_0(x) + \sum_{i=1}^m I_-(f_i(x)), \quad \text{subject to } Ax = b.$$

where $I_-(u) = 0$ if $u \leq 0$ and $= \infty$ if $u > 0$. We can approximate the indicator function by

$$\hat{I}_-(u) = -t^{-1} \log(-u), \quad t > 0$$

Logarithmic Barrier

Approximate the original problem by

$$\min t f_0(x) + \phi(x), \quad \text{subject to } Ax = b.$$

where $\phi(x) = -\sum_{i=1}^m \log(-f_i(x))$ is called the **logarithmic barrier**.

- **Central point**: the solution $x^*(t)$
- **Central path**: the set $\{x^*(t), t > 0\}$.
- The quality of the approximation improves as t grows. Let p^* be the optimal value of the original problem. Then

$$f_0(x^*(t)) - p^* \leq m/t$$

- When t is large, $t f_0 + \phi$ is difficult to minimize, since its Hessian varies rapidly near the boundary of the feasible set.
- Solve a sequence of problems, increasing t at each step

Barrier Method

At each iteration, compute $x^*(t)$ starting from the previously computed central point, and then increase t by a factor $\mu > 1$.

- Given a strictly feasible x , $t = t^{(0)} > 0$, $\mu > 0$, tolerance $\varepsilon > 0$
- Repeat the following steps
 - Compute $x^*(t)$ starting at x

$$\min t f_0(x) + \phi(x), \quad \text{subject to } Ax = b$$

- Update $x = x^*(t)$
- Stop if $m/t < \varepsilon$; otherwise, repeat the above steps with $t = \mu t$

It is recommended to choose μ between 10 to 20, and choose $t^{(0)}$ so that $m/t^{(0)}$ is the same order as $f_0(x^{(0)}) - p^*$.

Feasibility Problem

The barrier method requires a strictly feasible starting point $x^{(0)}$. Consider the phase I optimization problem if $x^{(0)}$ is unknown,

$$\min s, \quad \text{subject to } f_i(x) \leq s, \quad Ax = b$$

Let $\bar{p}^*$ be the optimal value,

- If $\bar{p}^* < 0$, the problem has a strictly feasible solution.
- If $\bar{p}^* > 0$, the problem is infeasible.
- If $\bar{p}^* = 0$ and the minimum is attained at x^* , the set of inequalities is feasible, but not strictly feasible. If $\bar{p}^* = 0$ and the minimum is not attained, the inequalities are infeasible.

In practice, treat $\bar{p}^* = 0$ if $|\bar{p}^*| < \varepsilon$ for some small ε .

KKT Conditions

A point x is equal to $x^*(t)$ if and only if there exists λ, v such that

$$\begin{aligned}\nabla f_0(x) + \sum_{i=1}^m \lambda_i \nabla f_i(x) + A^T v &= 0 \\ -\lambda_i f_i(x) &= t^{-1}, \quad i = 1, \dots, m \\ Ax = b, \quad f_i(x) &\leq 0, \quad i = 1, \dots, m \\ \lambda_i &\geq 0, \quad i = 1, \dots, m\end{aligned}$$

Let $f = (f_1, \dots, f_m)^T$ and $Df = \partial f / \partial x^T$ be the Jacobian. We can solve the equations for (x, λ, v) by Newton method.

$$\begin{pmatrix} \nabla f_0(x) + Df(x)^T \lambda + A^T v \\ -\text{diag}(\lambda) f(x) - t^{-1} 1_m \\ Ax - b \end{pmatrix} = 0$$

Primal-Dual Search Direction

The primal-dual search direction $\Delta y_{\text{pd}} = (\Delta x_{\text{pd}}, \Delta \lambda_{\text{pd}}, \Delta v_{\text{pd}})$ can be obtained by solving

$$\begin{pmatrix} \nabla^2 f_0(x) + \sum_i \lambda_i \nabla^2 f_i(x) & Df(x)^T & A^T \\ -\text{diag}(\lambda) Df(x) & -\text{diag}(f(x)) & 0 \\ A & 0 & 0 \end{pmatrix} \begin{pmatrix} \Delta x \\ \Delta \lambda \\ \Delta v \end{pmatrix} = - \begin{pmatrix} r_{\text{dual}} \\ r_{\text{cent}} \\ r_{\text{prim}} \end{pmatrix}$$

where r_{dual} , r_{cent} , and r_{prim} are dual residual, centrality residual, and primal residual, respectively,

$$r_{\text{dual}} = \nabla f_0(x) + Df(x)^T \lambda + A^T v$$

$$r_{\text{cent}} = -\text{diag}(\lambda) f(x) - t^{-1} \mathbf{1}_m$$

$$r_{\text{prim}} = Ax - b$$

Surrogate Duality Gap

In the primal-dual interior-point method, the iterates $x^{(k)}$, $\lambda^{(k)}$, and $v^{(k)}$ are not necessarily feasible, except in the limit as the algorithm converges.

The **surrogate duality gap** is

$$\hat{\eta}(x, \lambda) = -f(x)^T \lambda$$

It is the duality gap if x is primal feasible and λ, v are dual feasible, that is, $r_{\text{dual}} = 0$ and r_{prim} . The value of t corresponding to $\hat{\eta}$ is $m/\hat{\eta}$.

Primal-Dual Interior-Point Method

The goal is to find $y = (x, \lambda, v)$.

- Given x that satisfies $f_i(x) \leq 0$, $\lambda_i \geq 0$, $\mu > 1$, $\varepsilon_{\text{feas}} > 0$, $\varepsilon > 0$.
- Repeat the following steps.
 - Set $t = \mu m / \hat{\eta}$
 - Compute the primal-dual search direction Δy_{pd}
 - Determine step length $s > 0$ and set $y = y + s \Delta y_{\text{pd}}$
- Stop if $\|r_{\text{pri}}\|_2 \leq \varepsilon_{\text{feas}}$, $\|r_{\text{dual}}\|_2 \leq \varepsilon_{\text{feas}}$, and $\hat{\eta} \leq \varepsilon$

Linear Search

The step length can be determined by a backtracking line search.

- Compute the largest s that gives $\lambda \geq 0$,

$$s_{\max} = \sup\{s \in [0, 1] \mid \lambda + s\Delta\lambda \geq 0\}$$

- Start with $s = 0.99s_{\max}$ and backtrack $s = \beta s$ until we have

$$f(x + s\Delta x) \leq 0$$

- Continue multiplying s by β until we have

$$\|r(y + s\Delta y)\|_2 \leq (1 - \alpha s)\|r(y)\|_2$$

It is recommended to choose $\alpha \in [0.01, 0.1]$ and $\beta \in [0.3, 0.8]$.